

CHAT2App: Supporting Caregivers of Infants with Congenital Heart Disease

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ABSTRACT

Caregivers of post-operative infants treated for Congenital Heart Disease (CHD) undergo significant stress and anxiety following initial discharge. We examine the development of a mobile application to support caregivers and health care practitioners (HCPs) at this important time. Reflexive thematic analysis of responses to semi-structured interviews with both caregivers and HCPs is used to identify some of the key design requirements with a focus on addressing barriers and challenges to effective at home post-operative care-giving. We identify the need for careful and nuanced consideration of stake-holder perspectives and the unique challenges that this raises. These include effective recording and reminders for observations, clear communication channels and targeted and actionable outcomes to identifying symptoms of concern. A resulting prototype application is presented along with lessons learnt and remaining challenges uncovered during the study.

CCS CONCEPTS

• **Human-centered computing** → **Smartphones**; • **Applied computing** → **Health care information systems**.

KEYWORDS

healthcare, supporting caregivers, decision support

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1 INTRODUCTION

Congenital Heart Disease (CHD) refers to a variety of defects of the heart that may be present at birth. CHD has recently been estimated as occurring in 60.9 in every 10,000 births [7]. Approximately one-third of infants born with CHD will need urgent surgery in

infancy [8]. Infants with these complex conditions are at highest risk between the first and second operations, remaining fragile after their first surgery and in the early weeks after discharge home [6]. Caregivers are understandably anxious about taking their infant home and often worry that they will not know what to do or who to contact for advice.

The Congenital Heart Assessment Tool (CHAT) [9] was developed as a decision support tool for use by caregivers and community teams to identify and escalate early warning signs of deterioration in infants following discharge. CHAT uses a traffic light approach to determine an appropriate action for a caregiver or health care practitioner (HCP) (fig. 1). CHAT has undergone a significant evaluation as a paper based intervention. Evidence gained from studies about CHAT highlighted caregiver and HCP fears about discharging infants [11]. Additionally, barriers to the success of CHAT include the anxieties of caregivers in managing frequent monitoring and reliable transmission of data [23].

In order to address these barriers to the effective use of CHAT we conducted a study of HCPs and caregivers and use our findings to transform CHAT from a passive paper-based monitoring tool into a dynamic and responsive healthcare application. We provide a nuanced consideration of the information management challenge from both the perspective of the caregiver and HCP. We present a prototype application designed to address identified challenges in supporting caregivers of infants with CHD.

2 RELATED WORK

The challenges associated with care giving for an infant in collaboration with teams of HCP's has received attention not just from the healthcare management communities but also computer-supported cooperative work (CSCW) and human-computer interaction (HCI) communities. Health care professionals have a familiarity with the condition, how it impacts on the health of an infant and how it is likely to evolve over time. Coordination of care from a HCP perspective is often a matter of juggling roles and expectations of a variety of stake holders including other HCPs and caregivers [12]. Therefore, priority tends to focus on access to information, clear communication and workload management in order to support effective care decisions over the long term [1].

From the perspective of a caregiver, coordination of care is focused on being an arbiter of instantaneous information relating to the child and their care. The emphasis on any such intervention is

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Congenital Heart Assessment Tool 2 (CHAT2) Parent Version			
Behaviour	☐ Behaves normally e.g. content, smiles ☐ Stays awake or awakens quickly (as normal) ☐ Normal crying, easily soothed e.g. by feeding, comfort	☐ Quieter than usual ☐ Sleeping more than normal or less easy to settle ☐ Less responsive during normal activity ☐ Slightly irritable, unable to comfort	☐ Not responding to usual interactions ☐ Does not wake or if roused does not stay awake ☐ Weak or floppy ☐ High-pitched or continuous cry or no crying at all
Skin Colour Warmth	☐ Your baby's usual colour of skin, tongue, especially lips and nails ☐ Your baby's usual warmth of hands and feet ☐ Soft spot feels normal. ☐ Normal body temperature between 36.5 – 37.5 °C	☐ Blue tinge around mouth or finger tips, paler than usual, or skin is mottled ☐ Sweaty skin or cooler hands and feet than usual ☐ 'Soft spot' feels different to normal ☐ Temperature more than 37.5 – 38 °C or below 36 °C	☐ Very mottled, pale, blue or grey skin ☐ Very cold, sweaty or clammy ☐ Very puffy eyes, hands, feet or tummy ☐ Very sunken 'soft spot' ☐ Temperature more than 38 °C
Breathing	☐ Usual breathing rate and usual effort for your baby (ease of breathing) ☐ No cold symptoms, such as runny nose	☐ More breathless, faster or slower breathing rate ☐ Has cold symptoms, runny nose, cough, noisier breathing than normal ☐ Sucking in skin below ribs or in the lower front neck area, more than normal ☐ Nasal flaring (nostrils become wider as baby breathes)	☐ Not breathing ☐ Struggling to breathe or exhausted ☐ Cannot cry ☐ Very noisy breathing, grunting or gasping ☐ Severe 'sucking in skin' below ribs or lower front neck area, head is bobbing
Oxygen saturation	☐ Oxygen saturation range normal for your baby, if measured by you at home or by your community team ☐ Add individual details	☐ Oxygen saturation above or below the range for your baby on two separate readings, repeated 30 minutes apart	☐ If oxygen saturations constantly above or below the normal range for your baby ☐ Or you are unable to obtain an oxygen saturation reading
Feeding Nappies	☐ Baby feeding normally ☐ Regularly wet and dirty nappies ☐ Wakes for feeds and is hungry ☐ No vomits, unless small possets (milk coming back up after feed) are normal ☐ Weight increasing as expected: (Add individual)	☐ Any diarrhoea or vomiting, including vomit after medications or not keeping feeds down ☐ Struggling to feed, taking longer to feed, breathless on feeding, or sweaty during feeding ☐ Nappies not wet or dirty (change from usual nappy change times) ☐ Has lost weight or weight unchanged or more weight increase than expected	☐ Vomiting or diarrhoea much more than normal (or twice in a row if this is not normal) ☐ Cannot feed ☐ Dry nappies for 4 hours or more ☐ Blood in nappy
Parent response	☐ Parents/ carers happy with their baby	☐ Parents / carer concerned about their baby ☐ There may be no visible changes in the baby condition	☐ Extreme worry - Baby very unwell/not breathing ☐ Parents Start Basic Life Support assessment

Carry on normal care

Contact for Advice:
 Cardiac Ward (evening/night/weekend/bank holidays)
 Tel:
 Cardiac Nurse Specialist Team (9-5pm)
 Tel:

If you see any of these signs

 ring 999
 STRAIGHT AWAY

Figure 1: Congenital Heart Assessment Tool (CHAT) [10]

then to provide improvements on information management relating to records and events associated with care and the daily challenges of managing the condition [20, 21]. Work looking more holistically at the broader coordination of healthcare (including caregivers and HCPs) indicates a complex and fragmented approach is typical with wide disparity in expectation of time scales, loosely coupled planning approaches and disparate channels of communication [2].

Whilst the experience of caregivers and their interactions with HCPs is generally favourable, tensions do exist. Kientz [14] found that caregivers value the time they have with a HCP and are able to share large amounts of information about their infant but find it challenging when asked to recall specific details about their infant on the spot. Sepehri et al. [20] demonstrate that one of the most significant challenges faced by caregivers of an infant with significant health concerns is information management. Large numbers of documents and records quickly accumulate and communication with various HCPs can be disjointed and fragmented. Upon discharge, caregivers are also provided with an overwhelming amount of information and instructions. How effectively this information sharing is managed can have a significant impact on the success of treatment and the likelihood of readmission [3]. Overall, some clear themes emerge from the literature and these resonate with the findings of previous studies evaluating the effectiveness of CHAT [10, 23]

3 METHOD

The study used semi-structured, open-ended on-line video interviews to identify the needs and requirements of caregivers and HCPs when considering the use of a mobile application with a specific focus on the concerns outlined above. The focus of the interviews was kept broad enough to freely capture participant opinions and expectations without preconceptions (tables 1 & 2).

Participants were given information about the CHAT tool via the CHAT website [5] and a participant information sheet that explained the study and listed the semi-structured interview questions. The study was approved by the Ethics Committee of the University of Worcester and data was captured and processed in accordance with General Data Protection Regulation (GDPR) and UKRIO Code of Practice for Research. Participants were signposted to additional support services provided by the Little Hearts Matter network [16].

3.1 Participants

Participants were caregivers and HCPs recruited via the Congenital Cardiac Nurses Association (CCNA) social media [5] and snowballing. HCP participants were required to be in roles specialised in caring for infants with complex CHD. Nine HCPs took part across

What is your previous experience of home support for infants with complex cardiac conditions?
What experience do you have of using technology to monitor, document and communicate your own or an infant’s health status?
What do you think are the positive and negatives of using a mobile device to provide medical guidance rather than a paper-based method?
What barriers or challenges do you envisage in using a mobile device to monitor, document and communicate an infant’s health status?
How confident do you feel about recording information about your infant correctly?
How confident do you feel about recording information at the right time or frequency?
How confident do you feel about communicating with a healthcare professional when you need help.
What functionality do you think should be included in the application?
Would you consider using an application that actively notifies you when it’s time to monitor, document and communicate your infant’s health status?
How do you prefer to communicate with a health care professional? Telephone, text chat, video call or face-face appointment?

Table 1: Semi-Structured Caregivers Interview Schedule

What is your previous experience of supporting parents and carers at home with infants with complex cardiac conditions? Have you had any previous experience or knowledge of CHAT2?
What experience do you have of using technology to provide remote assistance and support to patients?
What do you think are the positive and negatives of using a mobile device to provide medical guidance on decision making rather than a paper-based method?
What barriers or challenges do you envisage in using a mobile device to monitor, document and communicate an infant’s health status?
How confident do you feel about the ability for parents or carers to record information about an infant correctly?
How confident do you feel about the ability of parents or carers to record information at the right time or frequency?
How confident do you feel about communicating with a parents or carers via a mobile application? How might this differ from a face-face consultation?
What functionality do you think should be included in the application?
Would you consider using an application that actively notifies you immediately when a parent or carer has raised a concern regarding the health status of an infant or would you prefer to check at specific times.
How do you prefer to communicate with parents or carers? Telephone, text chat, video call or face-face appointment?

Table 2: Semi-Structured HCP Interview Schedule

two online interviews representing five different health care organisations. Caregiver participants were required to have current or very recent experience of supporting an infant undergoing treatment for CHD. Four caregivers were interviewed across two online interview sessions. Due to the selection requirements participant numbers are relatively low for a study of this type. This was also exacerbated by a relatively low volunteer rate (approx 20%) due to the complexity and delicate nature of infant CHD cases.

3.2 Procedure

Caregivers were interviewed separately from the HCPs, with interview groups ranging from one to three participants. Interviews were with at least two researchers present to ensure monitoring for fair practice. Auto generated transcripts of the interviews were checked for accuracy against recordings of the interviews. Reflexive thematic analysis employing iterative refinement of theme codes [4] was then undertaken independently by two of the research team, who consolidated their findings into an agreed set of themes.

4 RESULTS

4.1 Parent Themes

The themes emerging from the caregivers’ observations fell naturally into three domains which might be used to inform the application requirements: carer situation and experience; perceived app uses and benefits, perceived disadvantages and technology opinions.

4.1.1 Carer situation and experience. Themes from caregivers discussing their carer situation and its impacts provide insight into the situations and emotions the caregivers are likely to be feeling when they are using the application. There was variation in the information and monitoring tools the caregivers reported receiving before discharge. A red flag warning leaflet was recalled by one parent, who stated: "Our cardiac nurse... gave us an A5 cardboard red flag thing that we’ve got in our kitchen. We’ve also given the nursery one" (P4). However, other caregivers did not recall receiving any red flag materials, although they recalled that the nurses verbally discussed the monitoring (P1, P2, P3). For one parent this led to feeling uncertain about the monitoring: "Two sets of information were being presented to me as one set of information and because it’s my first baby, I’m not sure what they’re telling me I need to look out specifically for because I have a baby with a heart condition" (P3).

Stress and fear were mentioned a few times. P1 stated that, even though the "training for home support was actually very good in terms of all of the feeding-tube training, the CPR training, the red flag training", with the apprehension of going home "I was like, please don’t make me go home with her." P2 recalled "when that much information flies at you, its like a wall flying at you". whilst P3 felt that "I think it’s quite emotional, isn’t it. So you kind of don’t take in everything". Caregivers felt such feelings had an adverse effect on the process of monitoring the babies: "You’re already feeling fear inside. So you actually forget." (P1); "Your brain, obviously everybody knows, is not there. So to then try and focus, that you’ve got to keep this [information], keep that, keep the other" (P2). The potential for an app to support with the monitoring and

flagging is therefore unsurprising, with all caregivers stating that they perceived the app as being a potentially useful tool.

4.1.2 Perceived application uses and benefits. Caregivers agreed that a mobile application would be useful, both at discharge and later: "I wish I'd had this [app] when we went home. This looks really good" (P2), "this would have been really, really useful even that much further on" (P1). As well as noting that the detail it would provide would be "really helpful" (P1), caregivers felt it would give them more confidence when discussing the baby with the GP: "this would have backed me up a little bit and given me that confidence to say to the GP 'No!'" (P2), "I think that [app] could really help me with the GP" (P1).

Providing reminders for checks and medication, were seen as important benefits by all the caregivers. There was discussion of expanding the application into a repository for recording data readings and timestamps: "I think the prompting [is important] because we definitely forgot to do sats on certain days at different points, and time stamping because I know the cardiac nurses would say, oh, when did you take that? And I'd be like 'Umm, it was Saturday or Sunday?'" (P3); "[Key advantage would be] prompting, you know... Have you done the feed? Ding! Have you done the meds? And Ding! Have you done your SATS monitoring?" (P1).

All caregivers proposed that the app might reduce the uncertainty of monitoring by providing clear examples to gauge measurements such as vomit quantities (P4), nappy contents (P2, P3), skin colour (P4) and breathing reception (P1,P2,P3,P4). The challenges of determining "normal" data for an individual child were summed up: "what does my baby normally look like?" (P1); "When it's your first baby you have nothing to compare to" (P2); "They're describing certain symptoms [but] you can't see them, you can't visualise them. So you kind of want to know what does normal breathing look like?" (P3)

The potential of the app to share information was raised by all the caregivers. Sharing information was seen as useful within families ("If you just tick like morning meds done...then your partner can see... Obviously we could just talk to each other but it's not always possible." (P1)), or with other HCPs, nurseries and carers (P2, P3, P4).

4.1.3 Perceived application disadvantages and technology opinions. A primary concern was that the app might become a replacement for practitioner contact: "I definitely wouldn't want the app to replace that direct contact with an actual healthcare professional" (P1); "One of the downsides of an app is if it seemed to replace weekly [meetings]" (P3).

Other concerns raised were: other carers might not have access (P2); potential interface problems and input screen errors (P3); too many apps (P1); too many app notifications (P3); having to keep the phone charged (P4), and potential problems from phone updates (P4). Also mentioned was the potential for the information to become inaccurate as the baby's condition progressed: "He's going to get worse and that's kind of that's normal for him at this point in time and it's kind of bad. Yeah. How do you factor that?" (P4).

4.2 HCP Themes

The HCP discussions highlighted themes representing similar domains to the caregivers.

4.2.1 Experience and practice. Although one HCP (H5) reported that CHAT was "not a tool that we regularly use", the other HCPs had either adopted a similar traffic light system for use by caregivers of babies (H2) or had already adopted the CHAT system (H6, H7, H8). HCPs familiar with CHAT reported caregivers finding it a useful and easy guide (H6), especially caregivers already engaging with the care teams: "We find that the parents that engage find it really useful and are really invested in it" (H8).

4.2.2 Perceived application uses and benefits. The application was seen as holding advantages over printed guides. Caregivers losing their paper guides was cited as a problem by H2 and H4. H2 reported they would be "more confident in [caregivers] using an app than the paper versions", perceiving the app as being easier to use and more likely to be consistently completed. Allowing the caregivers to set their own monitoring and prompt schedule was suggested by H2 as a likely method to improve compliance. H5 felt the app should provide a motivation reward "like a green check, like a 'well done, you've done your child's oxygen levels". Similarly, H1 cited the interactivity of the app as a motivator, "empowering the parents to identify a change and communicate that change ASAP appropriately".

The recording of information by the parent was seen by H3 as best restricted to saturation and weight, since "anything beyond that would be a bit tricky". However, monitoring feeding and hydration monitoring was seen as a useful addition in some cases, such as bouts of illness: "if you had the option of asking parents to put in that bit more information for a few days, that would be really useful clinically" (H2). Also mentioned as useful were recording parent contact with other medical departments (H1, H9) or showing that the prescription had changed (H8).

Similar to the caregivers, the HCPs proposed additional uses of the app. Providing additional information to the caregivers via Frequently Asked Questions (FAQs) was proposed by H5 and H9, such as providing information caregivers often requested from the team about topics such as medication replenishment. Video calls and using video to gauge normal behaviour (such as normal breathing) were proposed by H8 and H9: "I think there's a lot of power in them actually just taking a video of when their child is asleep and happy" (H8).

Extending the scope of the app to provide information sharing across the medical field, such as GPs, community nursing teams and A&E, or integrating the app into the electronic patient record (EPR), was proposed by the majority of the HCPs. Recommendations included storing care plans, discharge letters, readings or alerts for access by other medical teams (H2, H5, H6, H7, H9); taking images of wound recovery to share with doctors (H5); sharing medical information and emergency contact details amongst carers (H2); tracking the oxygen saturation of shunt dependent babies to determine whether they are outgrowing their shunts (H5); or, sharing alerts when a parent was skipping the monitoring (H2).

4.2.3 Perceived Application disadvantages and technology opinions. HCP concerns about the app focused mainly on practitioner

workload and process accountability. Although the potential for sharing information was strongly cited (as discussed above), it was felt that the app should ensure the onus to act on alerts should always be allocated to the parent. Staff unavailability, week-ends, sick-leave and out-of-hours alerts (H3, H4, H5, H6), and existing workloads including email volumes (H4, H8, H9), were cited regarding this. Potential confusion over tracking, where the parent might assume the medical team is constantly monitoring for alerts and responding to them, was also cited as a concern (H7). The possibility of caregivers misinterpreting measurements was raised by the HCPs. H8 pointed out that, although oxygen desaturation might be a natural consequence of the baby outgrowing their shunt, it might also be an indication of a virus infection. Thus: "the parents need to be aware that there might be other causes of desaturation, and that it's an important reason for them to get their children seen" (H8). Also discussed was the potential for over-measurement to lead to parental "heightened anxiety" (H5), and the possibility of the app overwhelming caregivers: "You're sending [parents] home with this sick baby and ... you're handing them basically a whole program that they need to try and navigate through" (H4).

4.2.4 Design Considerations. Regarding the design of the app, H8 noted that many residents in their constituency had Urdu or Punjabi as their primary language, so it would be useful if the app was usable by non-English readers. Experience with other apps drew HCPs to warn that the amount of text needed to be kept low (H2, H3), especially for caregivers in stress: "One of the things lots of our parents found quite overwhelming was the volume of text in the tool... Lots of parents just said ... 'When I'm panicking at home and I'm worried about my child, I will not be able to absorb that volume of text'" (H2). A need to adapt and customise measurements was noted, for example, to suit different babies or changes in one baby over time: "It would be really important to be able to adapt [the measurements] because we know that these children do tend to have a bit of a decline as they move towards their Glenn [heart surgery]" (H2); "Starting to outgrow their shunts, they need to be brought in for a pre-CPD shift, ...catheter and things, perhaps more frequent saturations at that point might be more useful" (H8); "The frequency [of measurements] would depend a little bit on the case that you're dealing with. ... and the same with the breathing rate. ... If you're making an app, it might need to be specific for that particular child" (H5).

5 SYSTEM DESIGN

In this section we discuss the design of the CHAT2App application, accounting for the findings from the caregiver and HCP interviews. An overview of the applications process flow is outlined in figure 2.

5.1 Monitoring Support: Journal Keeping

A key aspect of CHD care is identifying changes in behaviour and observations that indicate a deviation from normalcy. However, what is normal is unique to an infant and can change over time as an infant progresses. Both caregivers and HCP's identified normalcy of an observation as a particular challenge. However, correct interpretation of these logs requires specific medical knowledge and so we provide distinct views for both caregivers and HCPs to avoid misinterpretation.

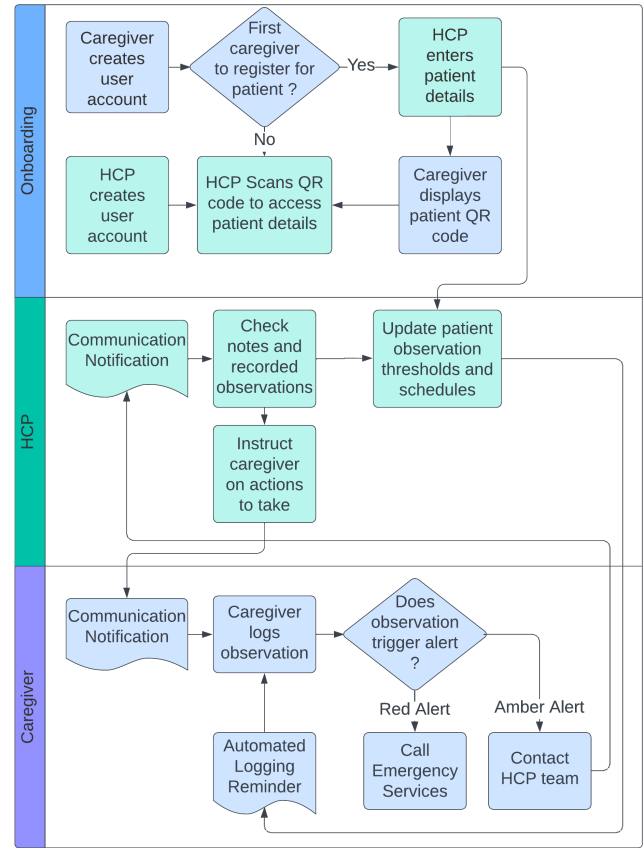
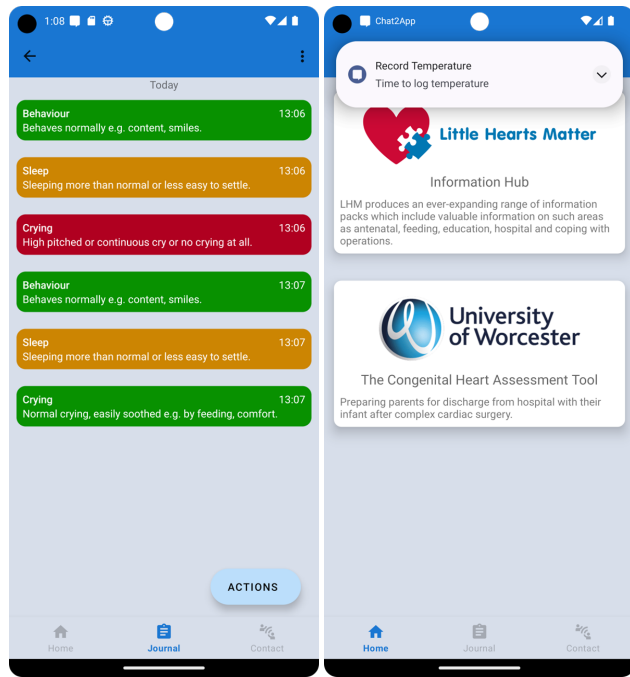


Figure 2: CHAT2App process flow

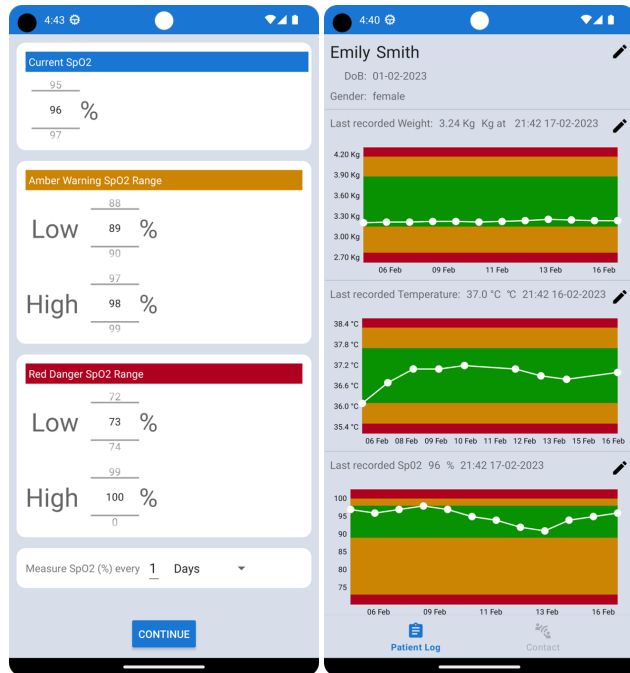
5.1.1 Journal Views. Caregivers see their recorded logs and a list of observations that are colour coded using CHAT (fig. 3a). This allows caregivers to get immediate feedback on normalcy with respect to that specific infant. However, this requires a HCP to set up appropriate thresholds for the infant to indicate suitable ranges for green, amber and red ranges for each parameter that is to be logged (fig. 4a). In order for a HCP to determine what is an appropriate range they will need to make a considered judgement based on the long term trends of the infant. For this reason the HCP team can see a more direct view of the longitudinal data in the form of an interactive chart where the long term trends and specific data points can be viewed and the threshold for green, amber and red status can be amended (fig. 4b). The chart supports interactive pinch zoom touch gestures to allow the user to adapt the range in both axis. Thresholds can be modified directly via the chart using an edit button.

5.1.2 Timely reminders of required observations. Caregivers and HCPs expressed concern about reliability and confidence in capturing observations that are accurate and timely. Both suggested that some prompting of caregivers to record information when required would be useful and that the schedule of these prompts should be adaptable to suit the infant. To this end the application utilises a notification system to prompt the caregiver to enter specific observations when required (fig. 3b). Whilst mobile device notifications



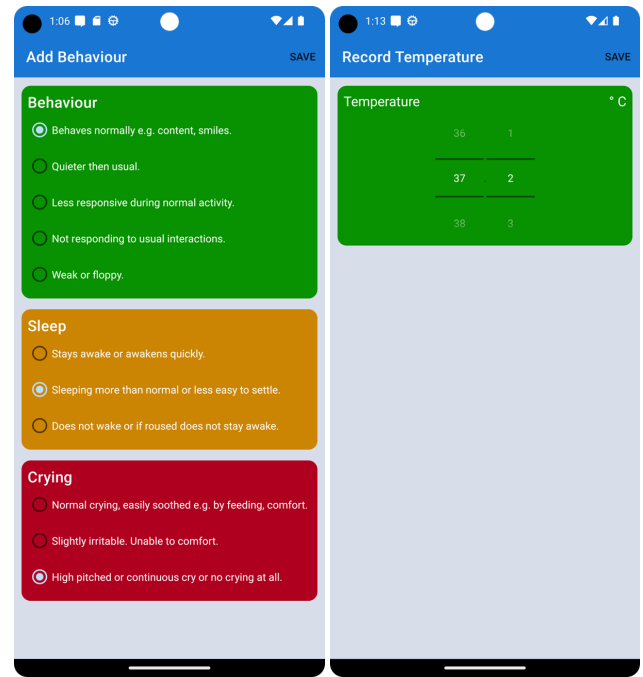
(a) Caregivers view of Logs (b) Notification as reminder to record observation

Figure 3: Caregiver Observation Monitoring



(a) Setting warning thresholds and scheduling reminder (b) HCP View of Logs

Figure 4: HCP Observation Monitoring



(a) Behavioural Observation Dialog (b) Numeric Data Entry Dialog

Figure 5: Logging Observations

are known to have some negative effects [19] these can be mitigated [24] by: (1) Ensuring notifications are persistent but not intrusive; (2) Generating new notifications only when required and without duplication; (3) Offering a clear and specific single purpose to the notification such as to log a particular observation; and (4) Allowing the scheduling to be adapted to suit the specific infant case.

To enable the schedule for these reminders to be personalised to the specific infant each observable metric can be uniquely set to notify at specific intervals by the HCP (fig. 4a) ensuring that notifications are only generated when required. These intervals can be adapted overtime by the HCP as the patient progresses. However, caregivers can override this schedule and directly enter observations at any time if they have immediate concerns regarding the appearance of behaviour of the infant. If this preempts a scheduled observation then that will remove the need to notify the user of the forthcoming scheduled observation.

Observations are recorded using simple form based response dialogues within minimal textual content. Behavioural observations are entered using radio button dialogues with short text descriptions of behaviour or appearance (fig. 5a). Numeric data is selected using touch enabled number pickers that are optimised for mobile interfaces with units and scales set appropriately for the required measurement to minimise data entry errors (fig. 5b). All observation recording dialogues used dynamic colour coding (green/amber/red) to ensure that caregivers are immediately aware of the alert status of a observation during data entry (fig. 5a).

5.2 Communication: Supporting an effective care team

Both caregivers and HCPs acknowledged the value of communication in the effective treatment of an infant. Both expressed the value in the use of video calling as a means to allow for visual observation of the infant whilst text messaging would allow a simple way to address less urgent queries. However, both agreed that these should not be seen as a replacement for face-face appointments. Communication was an area of some disparity between HCP and caregivers in our findings. Caregivers were concerned about obtaining timely responses. However, HCPs raised concerns about workload management associated with communicating with caregivers in an on-demand scenario. To this end the application supports two modes of communication:

5.2.1 Asynchronous Group Text Messaging. Asynchronous text messaging is supported by the application allowing anybody within the care team to message using text and images via a text messaging interface. The messages are shared across the whole team to ensure that everyone has access to all message content (fig. 6a). This approach provides a number of advantages over a live video call or face-face appointment [17]. The asynchronous nature means that carers and HCPs can raise questions and respond at a convenient time allowing accommodation of workload. It allows additional time to reflect on questions and review the infants medical history before responding [13]. It also provides an accessible and persistent record of the conversation for future reference.

5.2.2 Group Video Calling. Video call consultations are well suited to scenarios where there is a clear and immediate clinical goal such as diagnosis based on easy to describe symptoms [18]. Anybody within the care team can request a video call. The initiator of the request can specify who from the care team they would like to take part in the video call and can provide a brief note describing why the call is needed (fig. 6b). Once the request is received members of the team can choose to accept the request. Once appropriate team members have accepted the request a call can be scheduled for a specific time by any invitee or can be initiated immediately. This approach aims to address the concerns of caregivers by providing transparency about how a request is being handled and allows HCPs to manage workload associated with these requests.

5.3 Actionable Alerts

Our findings show that HCPs are very clear that responsibility for engaging with the CHAT principles (monitoring and requesting additional healthcare support) must lie with the caregivers in the first instance. Thus, the application utilises the same traffic light principles as outcomes of the decision making tool. However, since the application is running on a mobile device this affords the ability to ensure these outcomes are directly actionable by caregivers. The application supports this in the form of alert boxes on completion of observation logging. Where an observation warrants an amber warning then an appropriate alerts box is displayed (fig. 7a). The alert is persistent and once acknowledged by the user, navigates to the communications tools to allow the user to communicate their concerns with the HCP team directly. In the case of a red emergency alert, this requires immediate medical intervention so

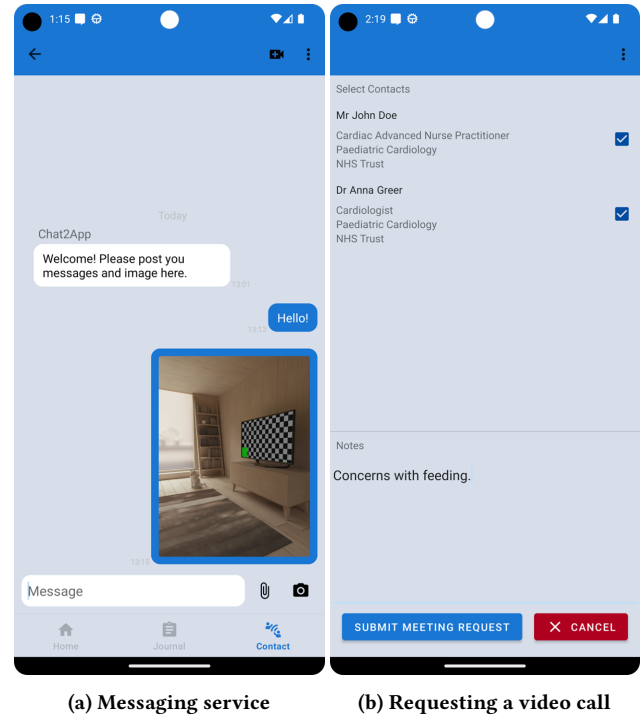


Figure 6: Communication

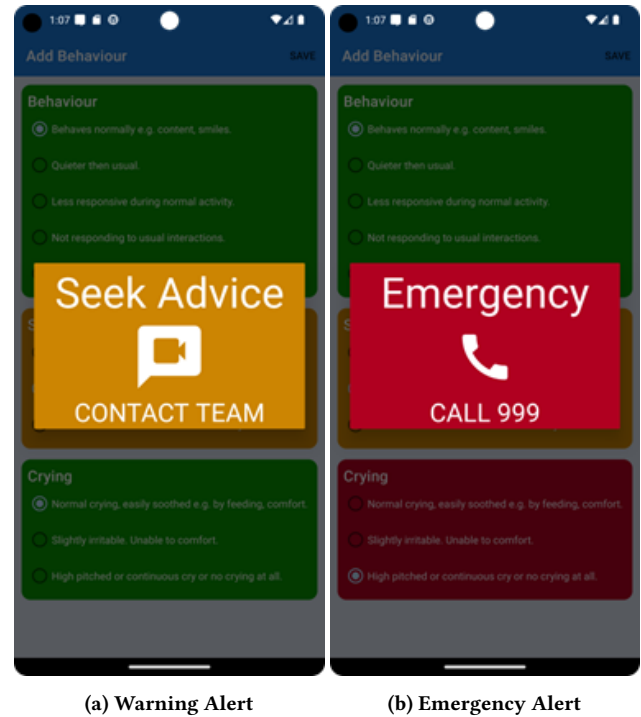


Figure 7: Actionable Alerts

triggers a persistent alert which, once acknowledged, triggers a call for medical assistance via the relevant emergency services (fig. 7b).

5.4 On-boarding

An effective on-boarding process is critical in ensuring that accounts are setup correctly and linked to the patient appropriately. There are unique on-boarding processes for caregivers and HCPs.

5.4.1 Initial caregiver setup. Initial setup of the application for the initial caregivers is expected to happen on a personal device in a hospital setting with HCP present. The initial caregiver creates a user account and provides some initial information about their infant. The applications will then instruct the caregiver to pass their device to the HCP who will ensure the correct initial observation parameters are setup for the infant including initial observations, schedule and alert thresholds (fig. 8a). The HCP will then demonstrate the functionality of the application to the caregivers present.

5.4.2 Linking HCPs. Once the initial caregiver is setup and an appropriate record for the infant has been generated then the HCP can add the infant to their list of patients for monitoring. This is done using a unique QR code generated by the caregiver application so that the HCP can scan the QR code to initiate the link (fig. 8b). HCPs may have multiple patients linked via the application.

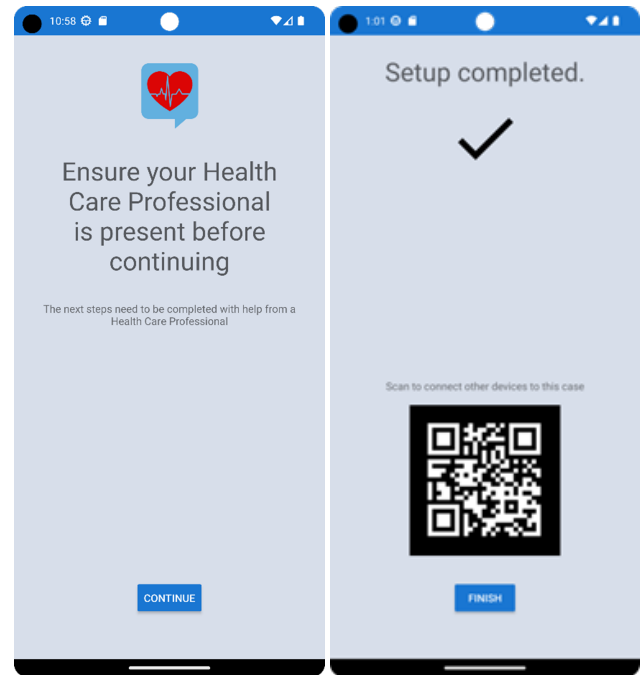
5.4.3 Adding and Transfer of caregiver role. Caregivers also talked about the overwhelming nature of being unable to easily include other trusted friends and family into the care team in an ad hoc fashion. This prevented caregivers from being able to take short breaks away from carer responsibilities. The application allows caregivers to invite new members securely using a unique QR code (fig. 8b). This allows family or friends to download the application and join the care team on a permanent or temporary basis to provide respite support.

6 DISCUSSION

Caregivers of post operative infants with CHD undergo an extremely challenging period of time when first discharged from hospital care. Whilst a passive decision support tool does help to address some aspects, many further challenges are apparent. Our research findings identify these challenges, many of which are quite tangible and addressable through the application of technology.

Information management: Whilst the challenges of information management in a health care setting are well recognised this work uncovers the impact that these issues have on caregivers and HCPs in a very specific scenario. Our application utilises logging that allows caregivers to easily record observations and visualisation tools that support caregivers and HCPs in decision-making tailored for their specific roles. This alleviates the burden of information sharing which is so apparent in the challenges experienced by caregivers in our study.

Decision support: Fundamentally, the application was built to enable improved decision support for caregivers. Identifying early warning symptoms and responding effectively is extremely important in improving the long term outcomes of an infant. The application simplifies the process of capturing observations and automates the process of classifying warning symptoms. Whilst this reduces some autonomy of the caregiver in the decision making process this was welcomed by both caregivers and HCP's as a positive step.



(a) HCP assists with initial on-boarding of caregivers (b) QR code approach to adding new caregivers and HCPs

Figure 8: On-boarding

Error reduction: Concern with making mistakes was common in the feedback from caregivers. Typically this related to two issues:

Normalcy - What is normal for an infant can seem subjective from the perspective of a caregiver with only experience of their own child. The role of a HCP in defining normalcy in an observable manner that a caregiver can readily comprehend is critical to addressing this concern. The application supports this through the ability of HCPs to adapt the observational thresholds for alerts and the use of video and images in communication.

Missed observations - Forgetting observations due to overwhelming workload associated with the care of an infant with significant medical needs is a tangible concern. The application addresses this through the use of a reminder notification system.

Care giving teams: Finally, our study uncovered some of the more nuanced communication challenges faced by caregivers and HCPs when forming a health care team. In particular the differing expectations relating to roles. Caregivers understandably want to be able to quickly gain support when the application identifies a concerning symptom. We found a variety of solutions used in practice but asynchronous text messaging was well suited in many cases as it allowed caregivers to raise pertinent queries whilst providing HCPs the flexibility to effectively respond within a managed time scale.

7 CONCLUSION

This work sets out with the aim of creating a mobile application that enhances the support for caregivers, and the HCPs supporting them, upon initial discharge of post operative infants with CHD from

hospital. We have analysed the concerns raised by both caregivers and HCPs in this context and have demonstrated that a technological solution can play a significant role in addressing many of those concerns. However, through this work we have identified the need for careful and nuanced consideration of differing roles and expectations of stake-holders and the unique challenges that this raises.

The next steps for this work are to evaluate the impact of this application when in use. This will provide an opportunity to address some findings from our study that were not directly addressed in our design and should be considered in further work.

Validation: Future work is also required to validate the application for use in a medical setting. This will require user evaluation of the application including quantitative data on user engagement and satisfaction. We will also need to quantify potential impacts on the accuracy and frequency of monitoring when compared to existing approaches.

Impact on Caregivers: The primary purpose of the application is to support caregivers in upon initial discharge of post operative infants with CHD from hospital. Whilst the desing of the application has been heavily influenced by the experience of caregivers it is important to confirm that the design as effectively achieved this my measuring improvements in montring performance but also anxiety among caregivers when compared with existing approaches.

Impact on HCP practice: HCPs expressed that, in practice, a lack of engagement with recording observations according to schedule is often used as a trigger for intervention in itself. Therefore, future versions of the application should report a trend of missed observations to HCPs. HCPs, especially those involved with nursing roles, demonstrated a clear concern for the impact of technology on workload. There is growing evidence that when technology is deployed without consideration of the role of the HCP it can often result in an increased rather than decreased workload [22]. Although some consideration for HCP workload have been made in the design of this application, further work should attempt to assess this impact directly [15].

We also acknowledge that future studies would ideally account for a broader range of participants from across diverse geographic regions and cultures.

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